

Root-Cause Investigation

Cause Level	Description	Failure Category	Evidence/Source
Immediate	Uncontrolled wide-open throttle; unable to apply the brakes to stop the car.	C4 Implementation / C2 Assumption	San Diego 911 call; crash telemetry. (Assumption: brakes would always physically overpower the engine).
Contributing	Lack of a Brake Override System (BOS). If both pedals are pressed, the system did not cut engine power.	C1 Requirements	System specifications lacked fail-safe priority for braking over acceleration.
Contributing	RTOS Task Death. The CPU could become overloaded, killing the fail-safe task that monitors throttle limits.	C3 Design / C4 Implementation	Barr Group source code analysis (10,000+ global variables, recursion, unchecked memory).
Root	Concealment of defects. Management prioritized cost savings and PR over investigating	C9 Management / C10 Ethics	DOJ Criminal Settlement detailing hidden memos and withheld engineering data.